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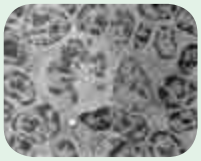
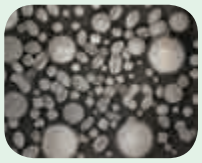
Comparison of Microscopes

Features	Light Microscope	Dark Field Microscope	Phase Contrast Microscope	Transmission Electron Microscope	Scanning Electron Microscope
Source of illumination for Image Formation	Visible light	Visible light	Visible light	Electrons	Electrons
Types of cells visualized	Individual cells can be visualised, even living ones.	Individual cells can be visualised, even living ones.	Individual cells can be visualised, even living ones.	Thin sections of the specimen are obtained. The electron beam pass through the sections and form an image with high magnification and high resolution.	The specimen is coated with gold and the electrons are reflected back and give the details of surface topography of the specimen.
Image	2-D	2-D	2-D	2-D	3-D
Nature of Lenses	Glass lenses	Glass lenses	Glass lenses	One electrostatic lens with few electromagnetic lenses	One electrostatic lens with few electromagnetic lenses
Medium	Air/oil	Air/oil	Air/oil	Vacuum	Vacuum
Specimen mounting	Glass slides	Glass slides	Glass slides	Mounted on coated or uncoated copper grids	Mounted on aluminium stubs and are coated in gold
Focusing and Magnification Adjustments	Changing objectives	Changing objectives	Changing objectives	Electrical, using deflection coil	Electrical, using deflection coil
Means for obtaining specimen Contrast	Light diffraction	Through patch stop	Through phase plate	Electron scattering	Electron scattering
Microscope picture